1) a) Binary digits

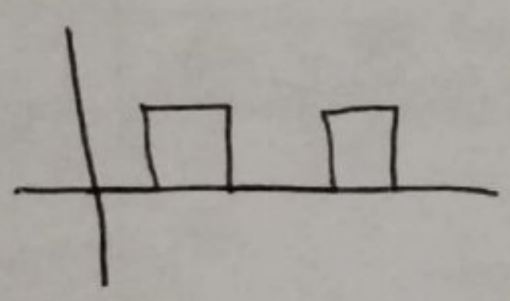
Binary digit refers to any digital encoding / decoding system in which there are exactly two possible state.

Logical levels

One of a finite number of states that a digital signal can inhabit.

Digital waveform

A waveform that switchs between two states (0/1 or high/low)



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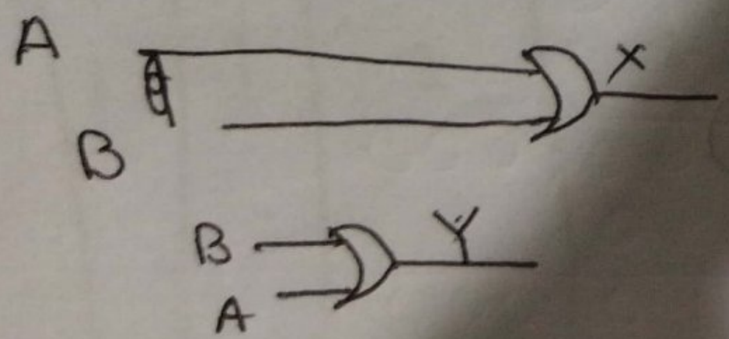
d) Boolean algebra is the branch of algebra in which ~~vafe~~ values of variables are the truth values: true and false.

Laws of Boolean algebra

1) Commulative law

$$A+B = B+A$$

$$AB = BA$$



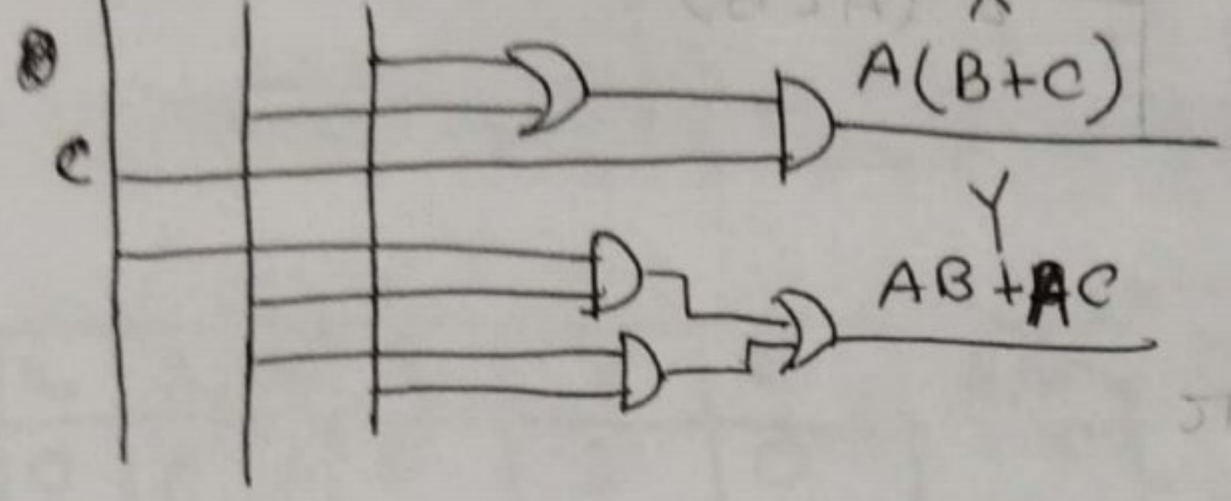
A	B	A+B	B+A	AB	BA
0	1	1	1	0	0
1	0	1	1	0	0
0	0	0	0	0	0
1	1	1	1	1	1

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ii) Distributive

$$A(B+C) = AB+AC$$



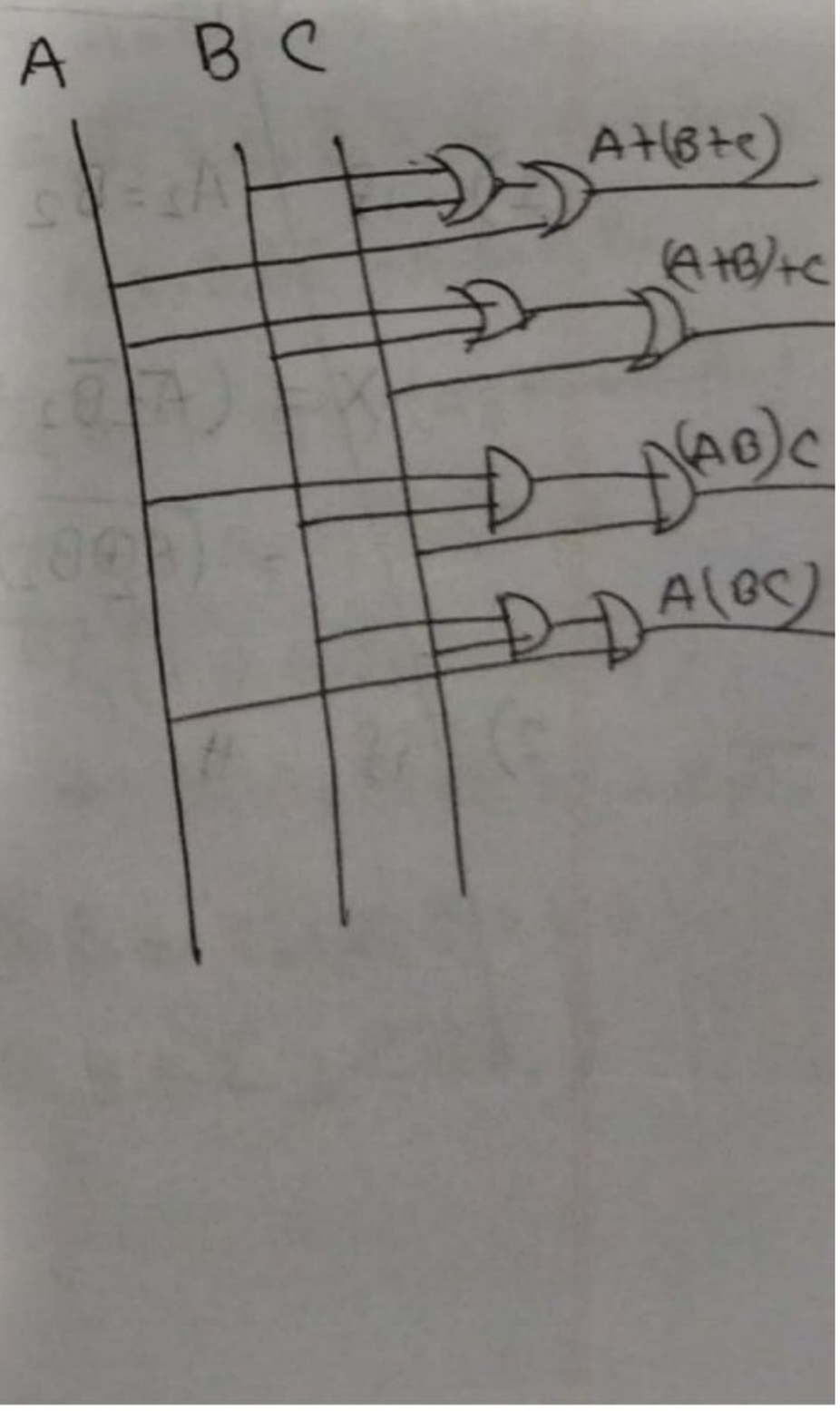
A	B	C	X	Y
0	0	0	0	0
0	0	1	0	0
0	0	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

iii) Associative law

$$A+(B+C) = (A+B)+C$$

$$A(BC) = (AB)C$$

A	B	C	A+(B+C)	(A+B)+C	A(BC)	(AB)C
0	0	0	0	0	0	0
0	0	1	1	1	0	0
0	1	0	1	1	0	0
0	1	1	1	1	0	0
1	0	0	1	1	0	0
1	0	1	1	1	0	0
1	1	0	1	1	0	0
1	1	1	1	1	1	1



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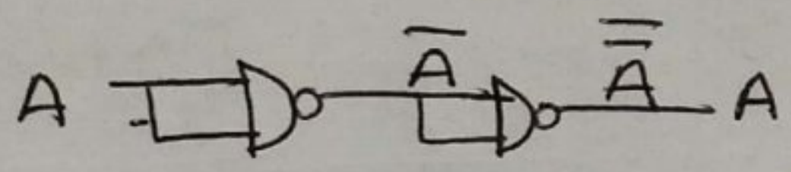
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c) NAND, NOR:

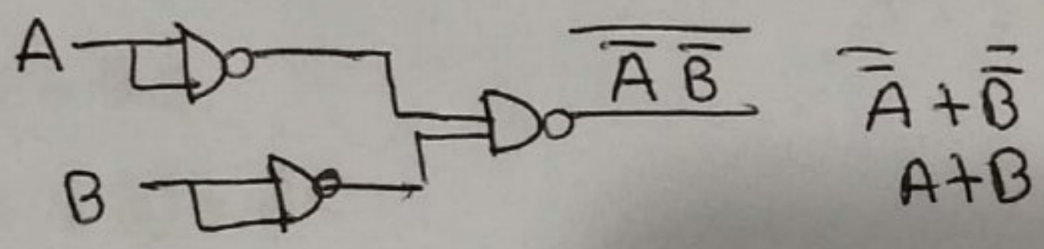
Cause we can implement AND, OR and NOT function from this gates.

NAND as universal ~~0~~ gate –

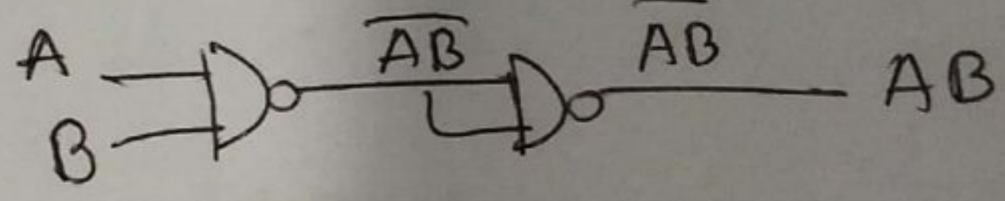
Not →



OR →



AND →



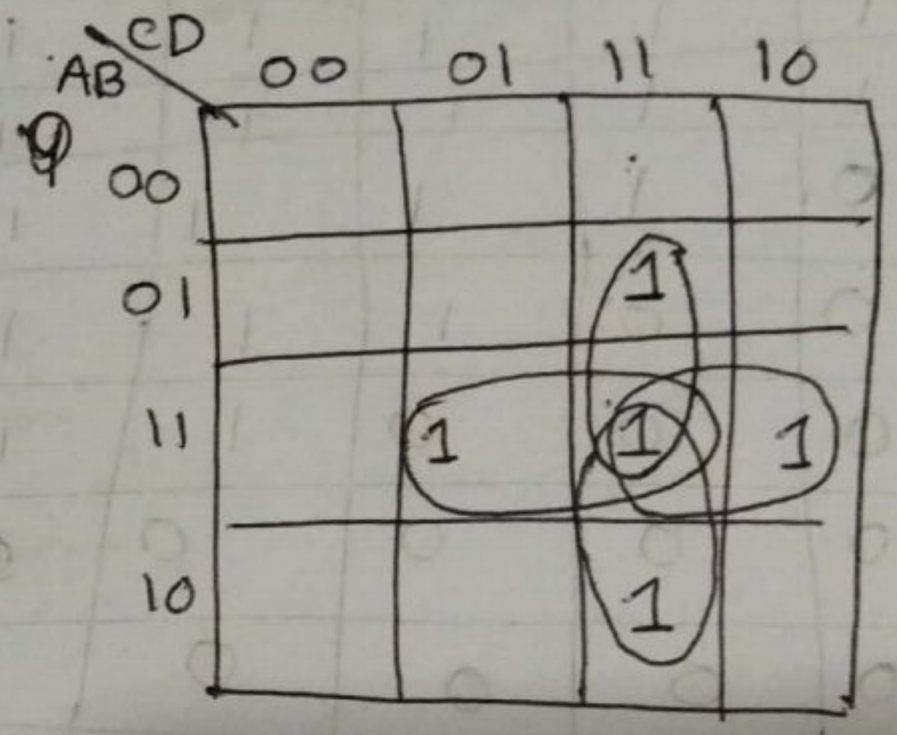
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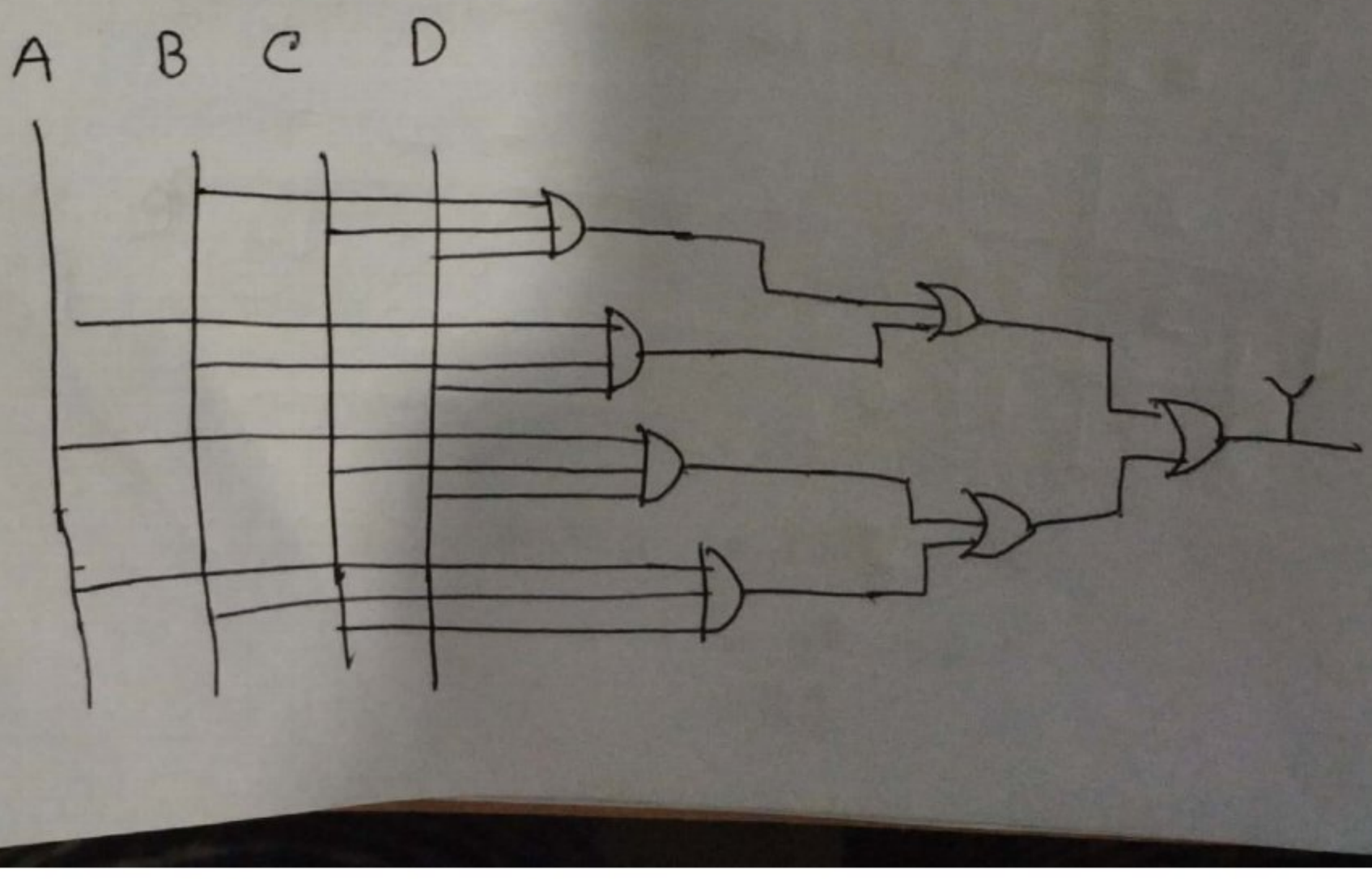
2) a)

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

$Y = \bar{A}BCD + A\bar{B}CD + AB\bar{C}D + ABC\bar{D} + ABCD$



$Y = B\bar{C}D + ABD + A\bar{C}D + ABC$



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b) i) $\overline{\overline{A+B\bar{C}}+D(\overline{E+\bar{F}})}$

$= \overline{\overline{A+B\bar{C}}} \cdot \overline{D(\overline{E+\bar{F}})}$

$= (A+B\bar{C})\,(\bar{D}+\overline{(E+\bar{F})})$

$= (A+B\bar{C})\,(\bar{D}+E+\bar{F})$

ii) $\overline{\overline{ABC}+D+F} = \overline{\overline{ABC}} \cdot \overline{D+F}$

$= \overline{\overline{ABC}} \cdot \bar{D}\ \bar{F}$

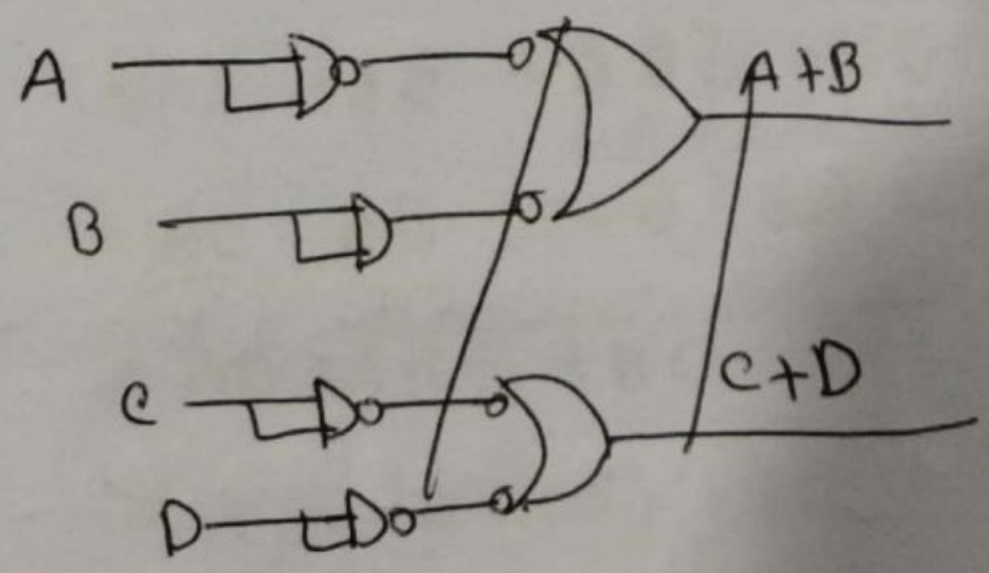
$= ABC\bar{D}\bar{F}$

c) 1010111 → 1100101

XOR

A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

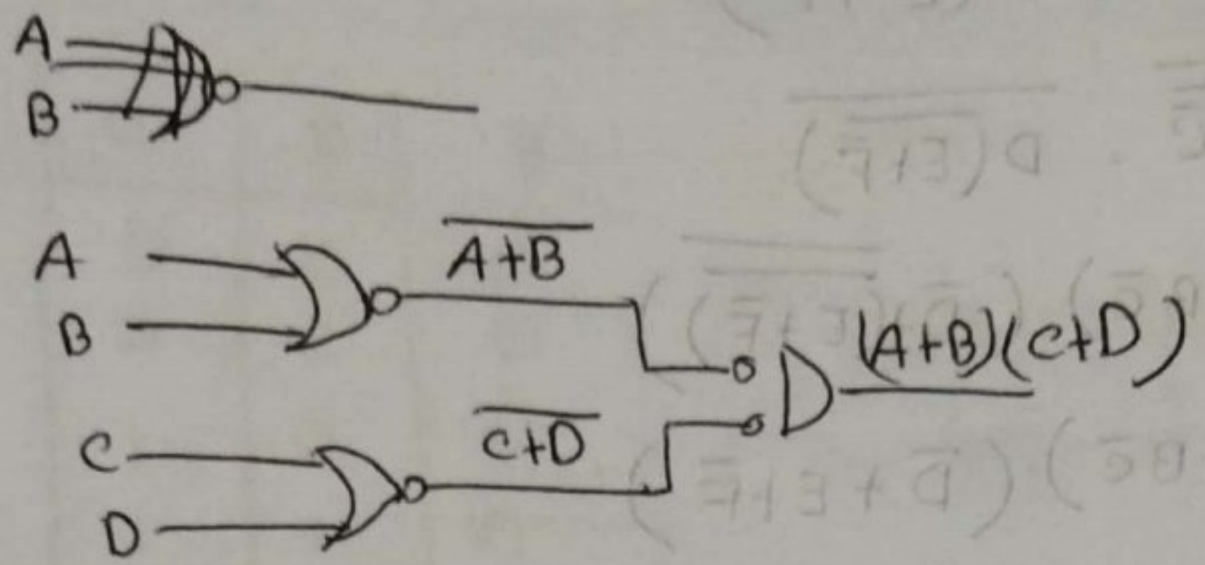
d) ~~$X = (A+B)(C+D)$~~



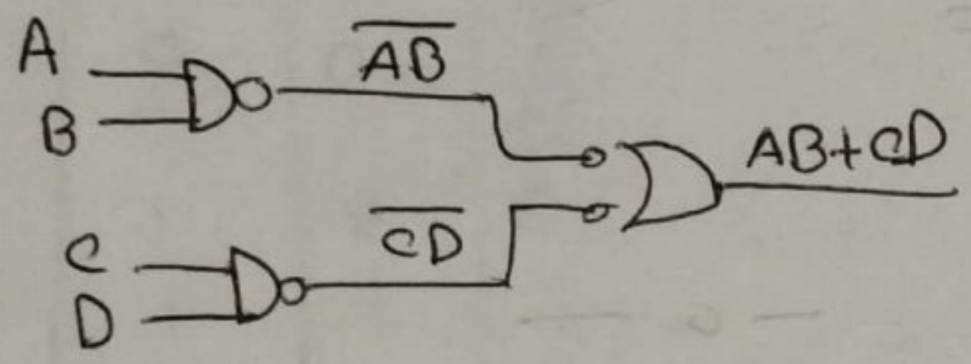
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d) $X = (A+B)(C+D)$



$X = AB + CD$



3) a) $\bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C + ABC$

$= BC(A+\bar{A}) + \bar{B}\bar{C}(A+\bar{A}) + \bar{A}BC$

$= BC + \bar{B}\bar{C} + \bar{A}BC$

$= BC(1+\bar{A}) + \bar{B}\bar{C}$

$= BC + \bar{B}\bar{C}$

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3) b) SOP → $AB + \bar{A}B + AB\bar{C} + ABC$

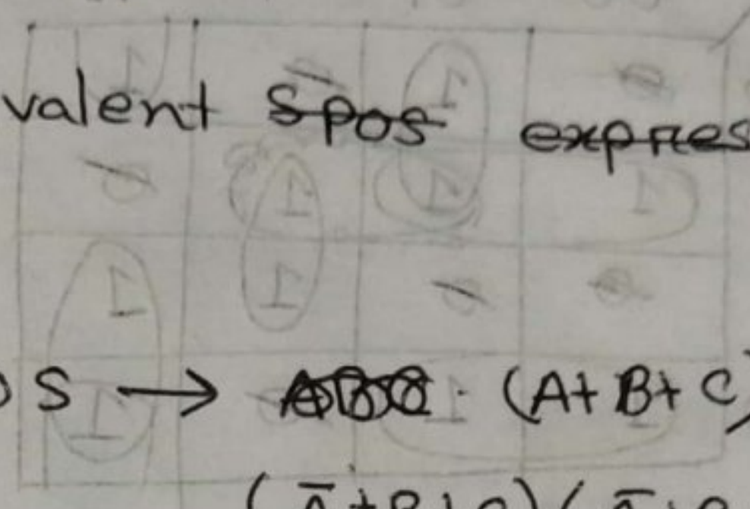
SSOP → $ABC + AB\bar{C} + \bar{A}BC + \bar{A}B\bar{C}$

$\frac{111}{7}$ $\frac{110}{6}$ $\frac{011}{3}$ $\frac{010}{2}$

So 0, 1, 4, 5 are missing

So ~~equal~~ equivalent ~~SPOS~~ ~~expression~~

			Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



SPOS → ~~ABC~~ $(A+B+C)(A+B+\bar{C})(\bar{A}+B+C)(\bar{A}+B+\bar{C})$

— o —

a) $\bar{A}BC + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C + ABC$

$= BC(A+\bar{A}) + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC$

$= BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}BC$

$= BC + A\bar{B}(C+\bar{C}) + \bar{A}\bar{B}\bar{C}$

$= BC + A\bar{B} + \bar{A}\bar{B}\bar{C}$

$= A\bar{B} + \bar{A}\bar{B}\bar{C} + BC$

$= A\bar{B} + \bar{B}(A+\bar{A}\bar{C}) + BC$ [$A + \bar{A}C = A + C$]

$= \bar{B}(A+\bar{C}) + BC$

$= A\bar{B} + \bar{B}\bar{C} + BC$

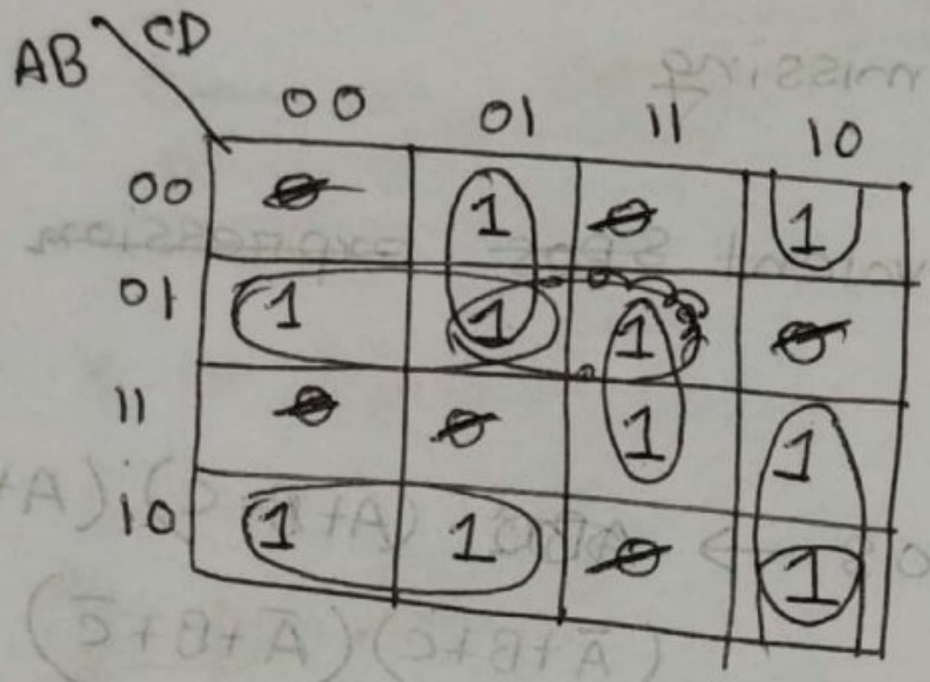
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c) $\bar{A}\bar{B}\bar{C}D + A\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}D + A\bar{B}C\bar{D} + \bar{A}BCD + ABC\bar{D}$

SOP → A=1, $\bar{A}$=0 $+\bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + ABCD$

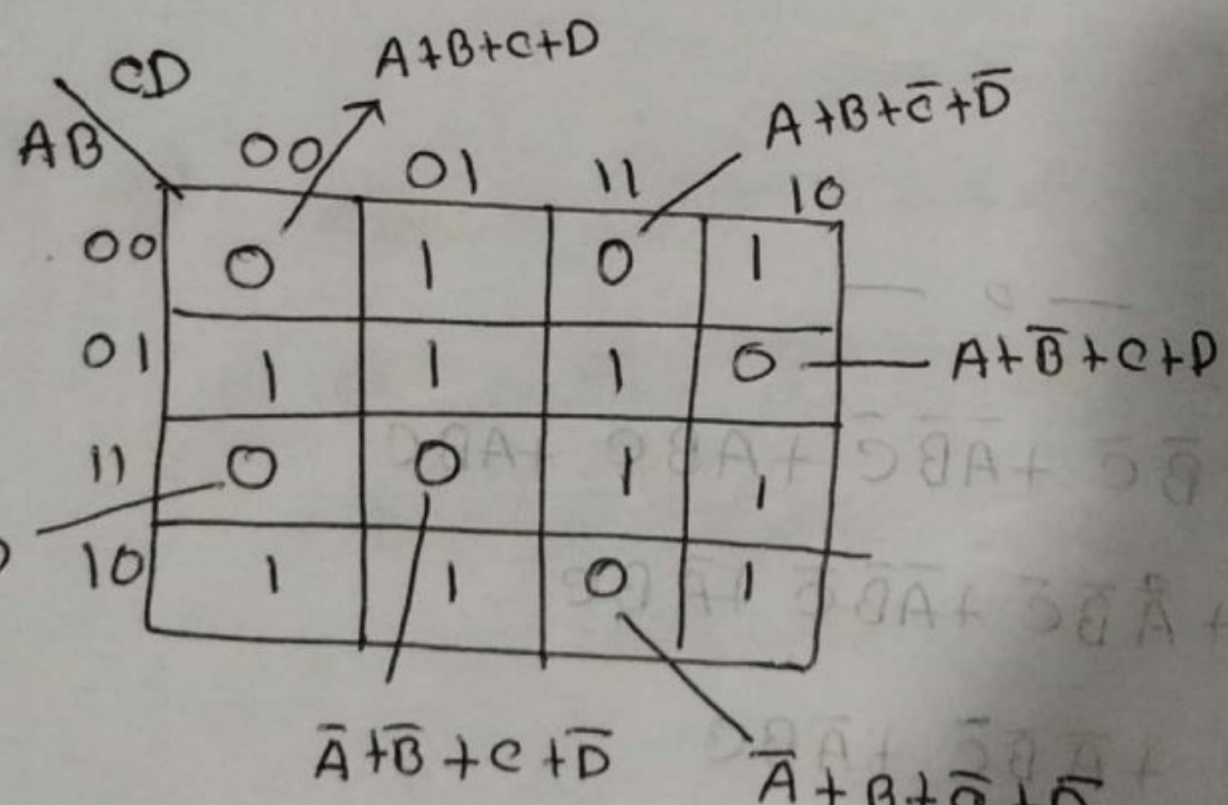


SOP Y

minimum SOP → $\bar{A}\bar{C}D + \bar{A}B\bar{C} +$ ~~$A\bar{B}D$~~

$BCD + AC\bar{D} + A\bar{B}\bar{C} + \bar{B}C\bar{D}$

POS A=0, $\bar{A}$=1



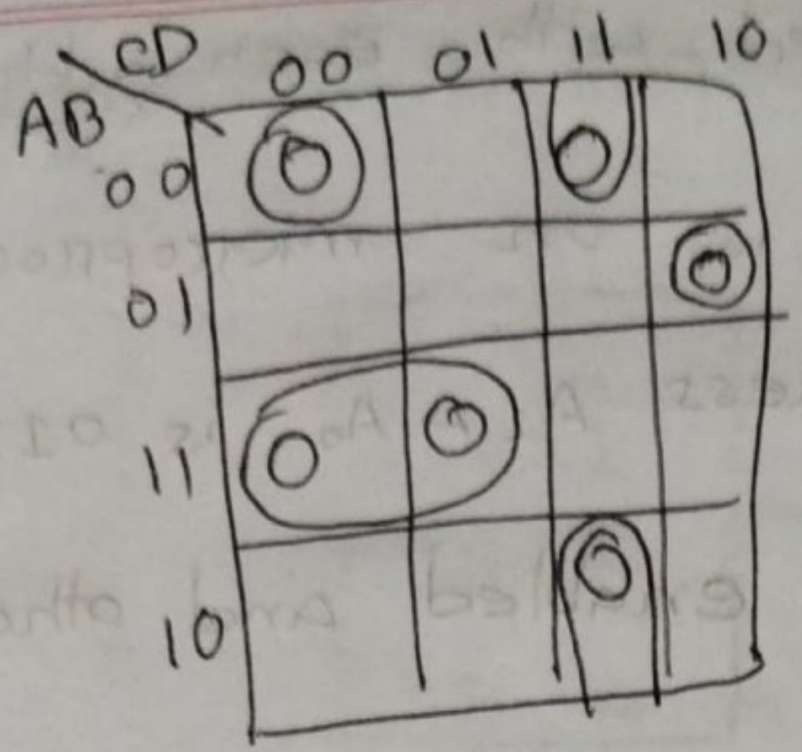
SPOS → $(A+B+C+D)(\bar{A}+\bar{B}+C+\bar{D})$

$(A+B+\bar{C}+\bar{D})(\bar{A}+B+\bar{C}+\bar{D})(A+\bar{B}+C+D)$

$(\bar{A}+\bar{B}+C+D)$

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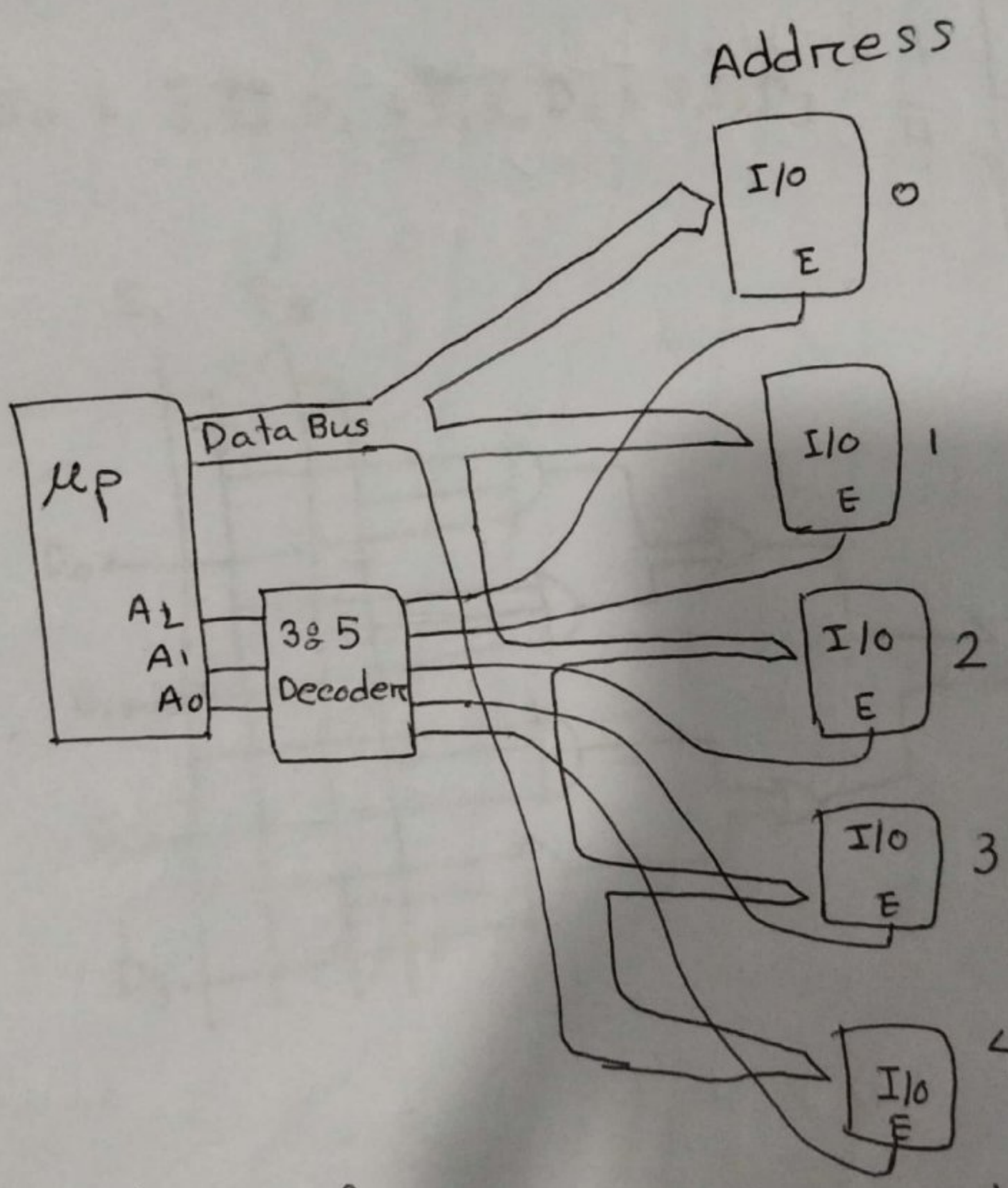
Minimum POS

$= (A+B+C+D)(A+\bar{B}+\bar{C}+D)$

$(B+\bar{C}+\bar{D})(\bar{A}+\bar{B}+C)$

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4) a)



~~Guess~~ to Now for the five address microprocessor ~~the~~. use 3:5 decoders to select the addresses. The output of 3:5

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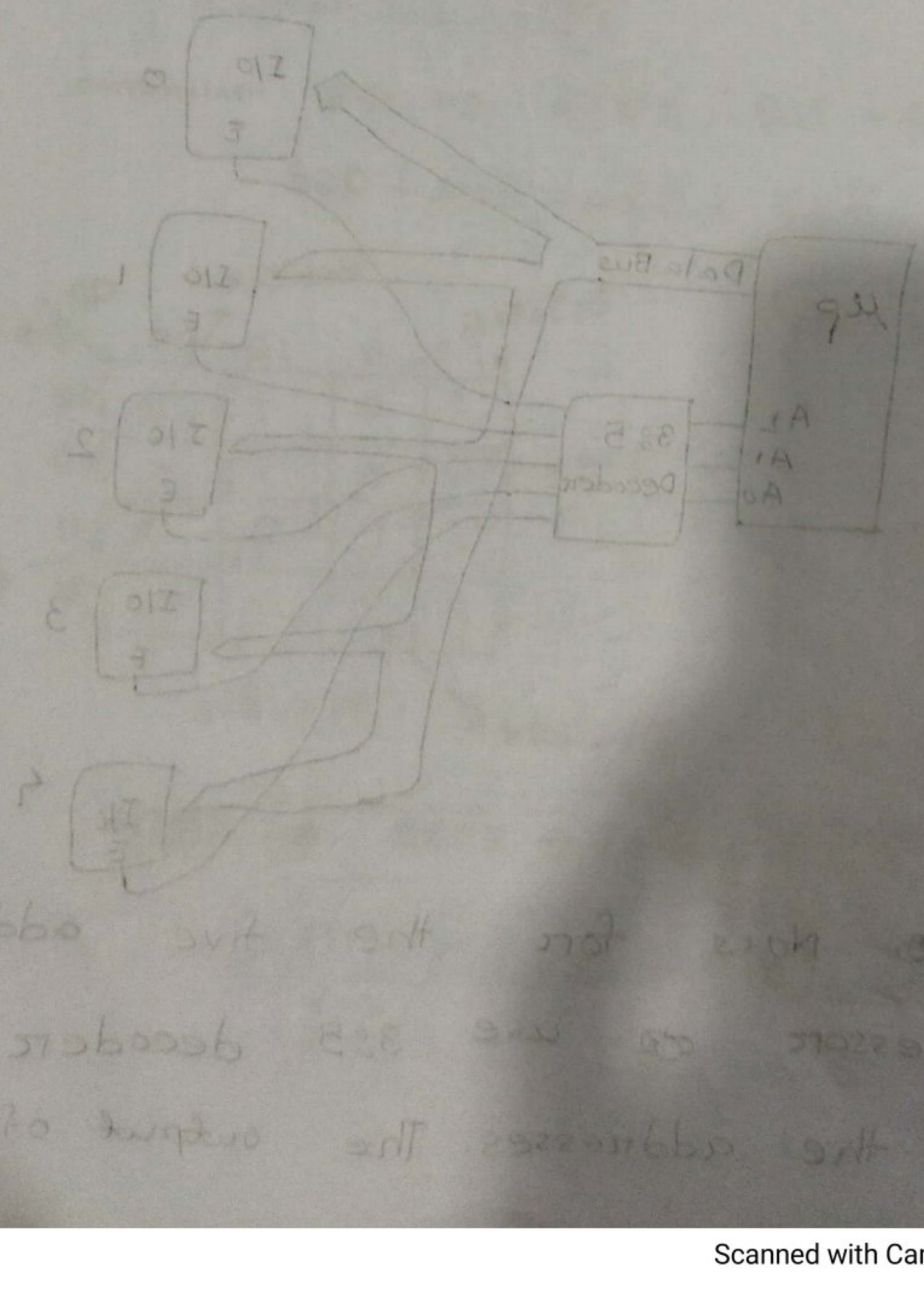
decoder is connected with each address

First the address is on microprocess via address bus. Guess $A_2 A_1 A_0$ is 011.

So Address 3 is enabled and others are deactive.

— o —

Ans

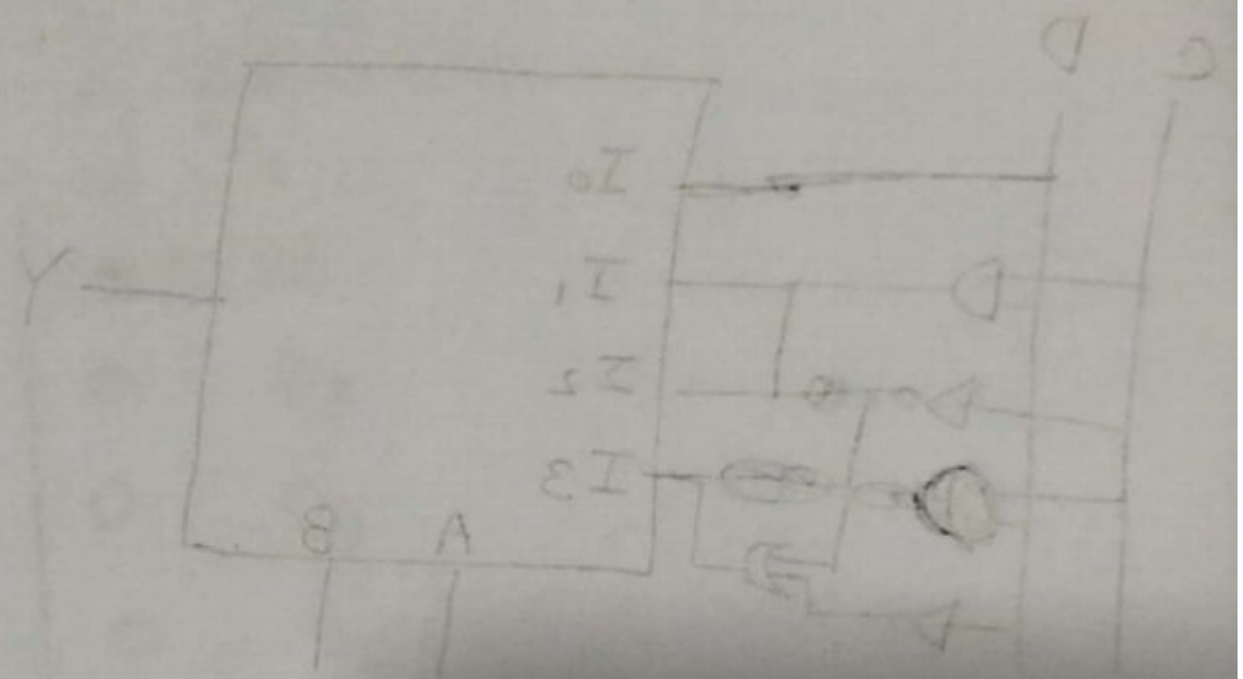


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4) b) $A\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}$

A	B	C	Y
0	0	0	1 D_0
0	0	1	1 D_1
0	1	0	1 D_2
0	1	1	1 D_3
1	0	0	0 D_4
1	0	1	1 D_5
1	1	0	0 D_6
1	1	1	0 D_7



In 4x1 MUX $2^2 = 2$ select line.

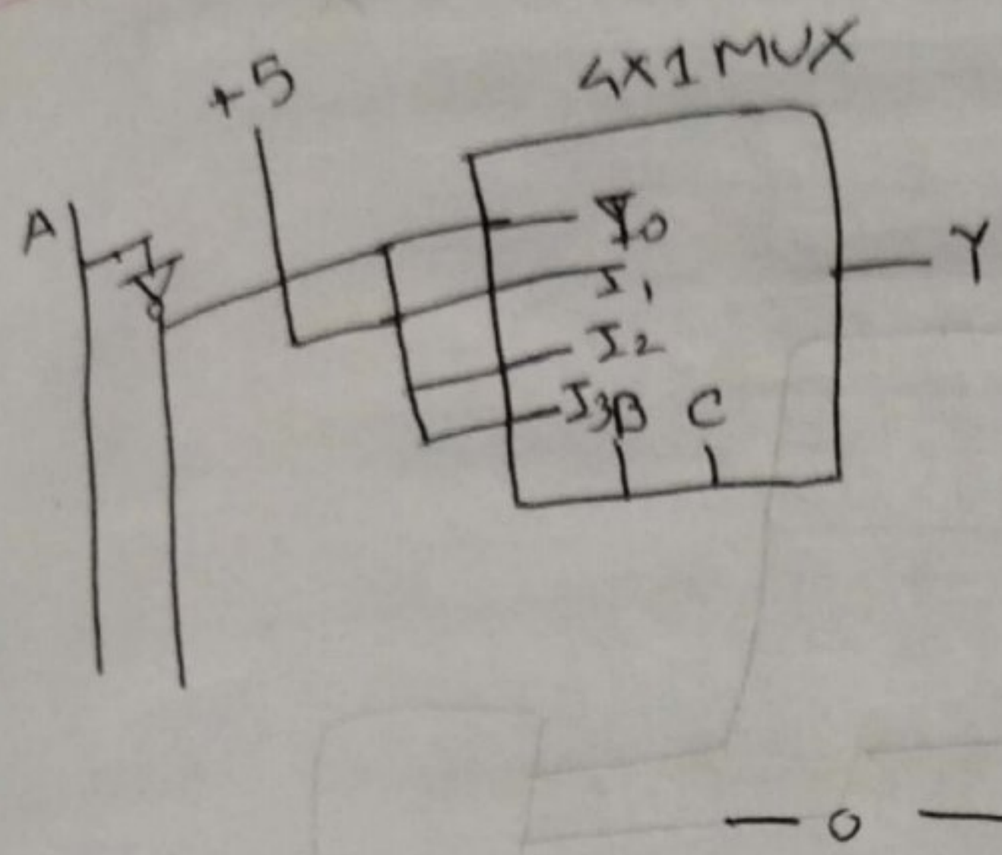
let A → input line

B, C → select line

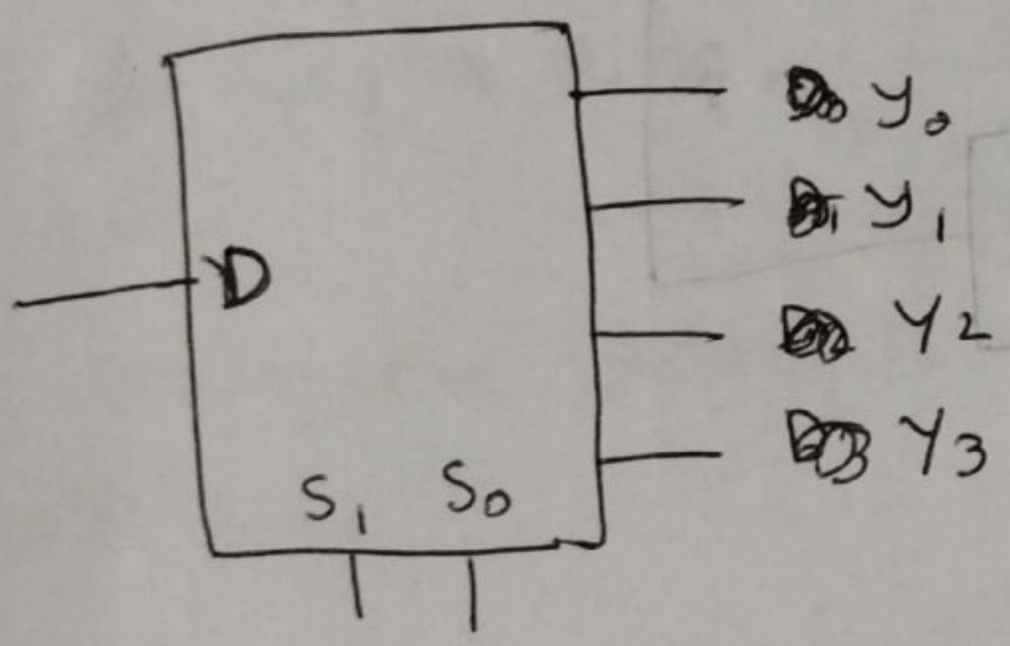
	I_0	I_1	I_2	I_3
$\bar{A}$	⓪	①	②	③
A	4	⑤	6	7
	$\bar{A}$	1	$\bar{A}$	$\bar{A}$

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c) 1x4 DeMUX



S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	D
0	1	0	0	D	0
1	0	0	D	0	0
1	1	D	0	0	0

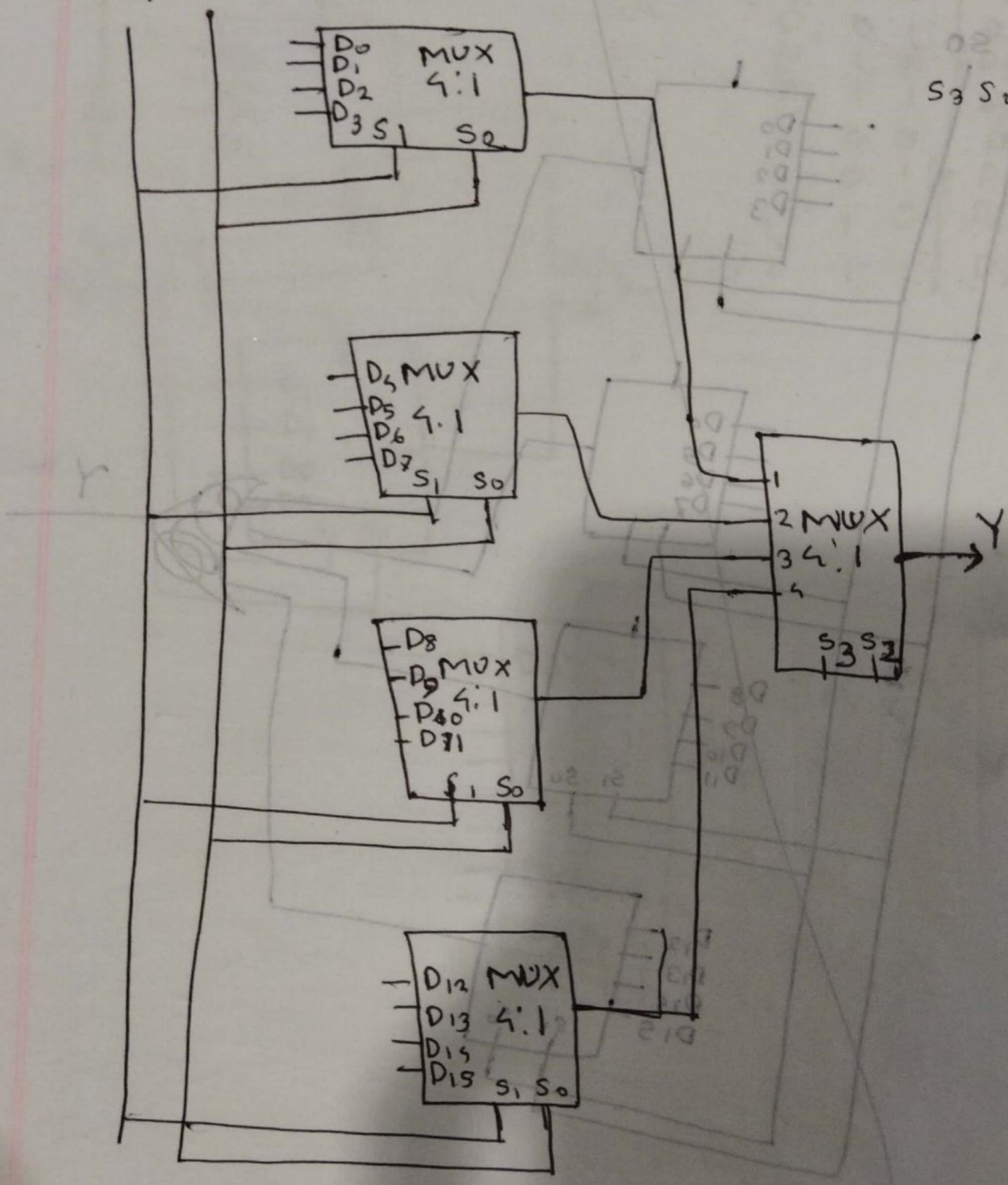
$$Y_0 = \overline{S_1}\,\overline{S_0}\,D$$
$$Y_1 = \overline{S_1}\,S_0\,D$$
$$Y_2 = S_1\,\overline{S_0}\,D$$
$$Y_3 = S_1 S_0\,D$$

S_1	S_0	D_0	P_1	P_2	P_3
0	0	1			
0	1		1		
1	0			1	
1	1				1

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16:1 MUX using 4:1 MUX



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5) a) 3 bit up down synchronous counter

clk	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

down

UP

b) for up

$J_0 = 1 = k_0$

$J_1 = k_1 = Q_0$

$J_2 = k_2 = Q_0 Q_1$

for down

$J_0 = k_0 = 1$

$J_1 = k_1 = \overline{Q_0}$

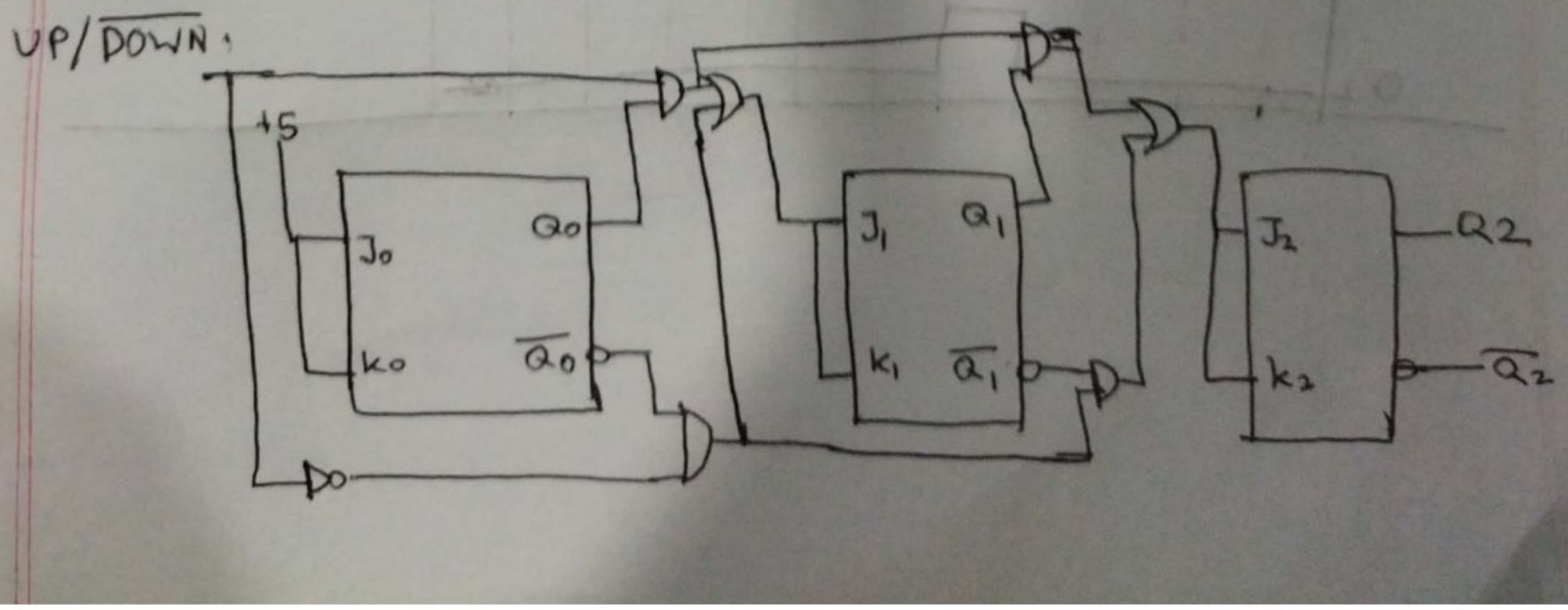
$J_2 = k_2 = \overline{Q_0}\,\overline{Q_1}$

the equation for up - down -

$J_0 = k_0 = 1$

$J_1 = k_1 = Q_0 UP + \overline{Q_0}\,\overline{down}$

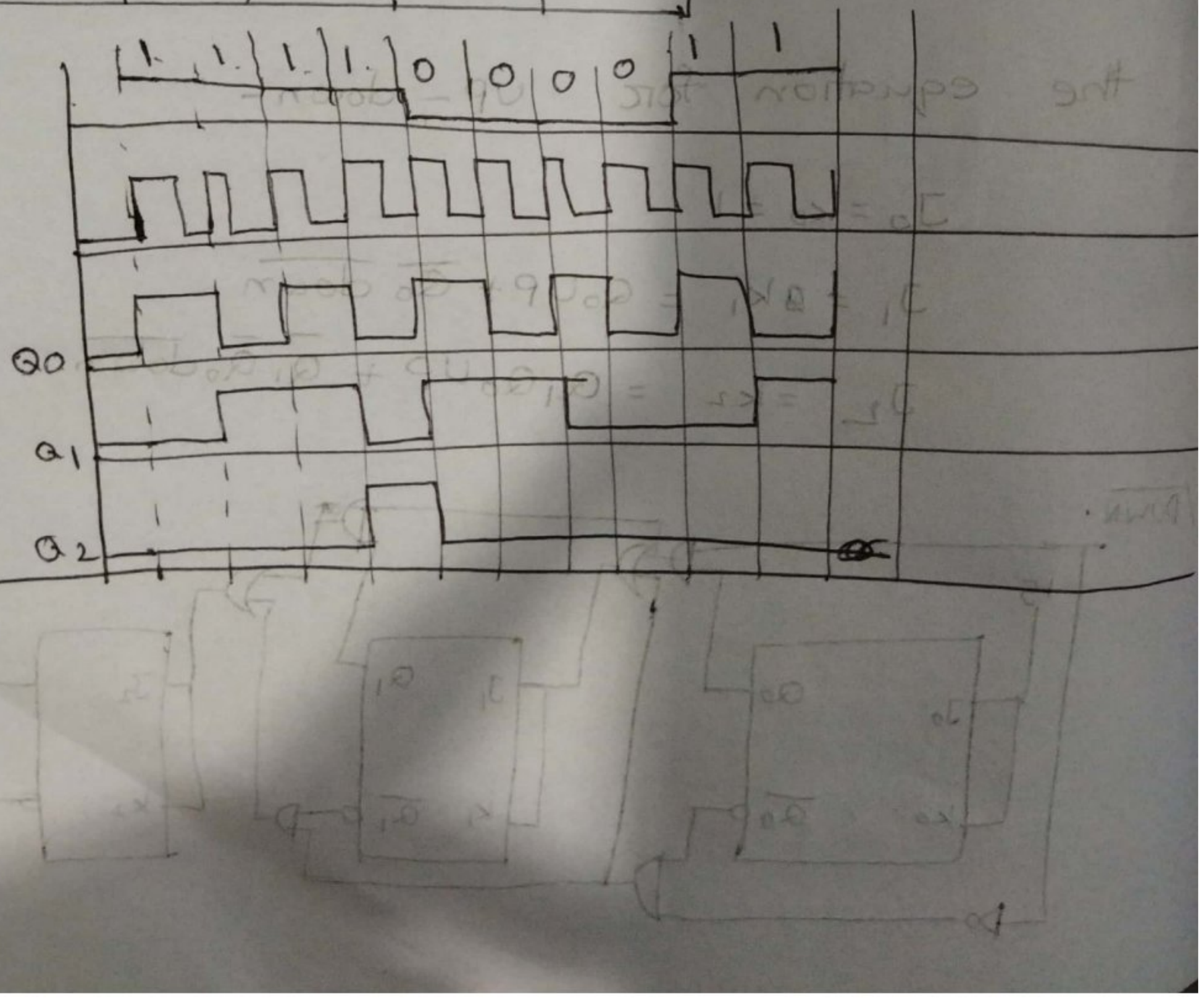
$J_2 = k_2 = Q_1 Q_0 UP + \overline{Q_1}\,\overline{Q_0}\,\overline{down}$



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U/D	CLK	Q_2	Q_1	Q_0
·	0	0	0	0
1	1	0	0	1
1	2	0	1	0
1	3	0	1	1
1	4	1	0	0
0	5	0	1	1
0	6	0	1	0
0	7	0	0	1
0	8	0	0	0
1	9	0	0	1
1	10	0	1	0

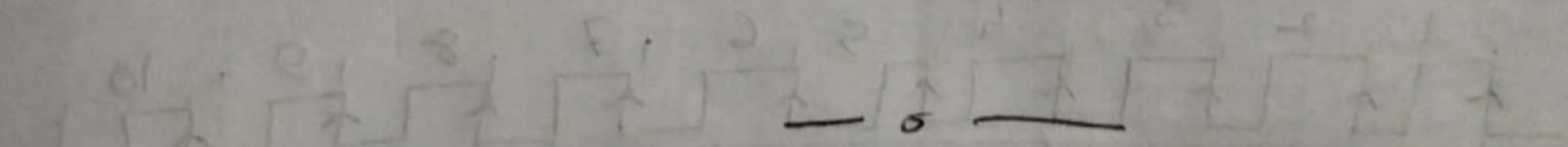


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5) b)

Synchronous	Asynchronous
1) All flipflop are triggered with same clock	1) Different clock
2) Faster in operation	2) Slower
3) Doesn't produce decoding error	3) Produce decoding error.
4) known as parallel counter.	4) Serial counter.
5) Example → Ring, Jonshon	5) Exam- Ripple up, down
6) propagation delay is less.	6) Propagation delay is high.



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Mod-10 asynchronous counter

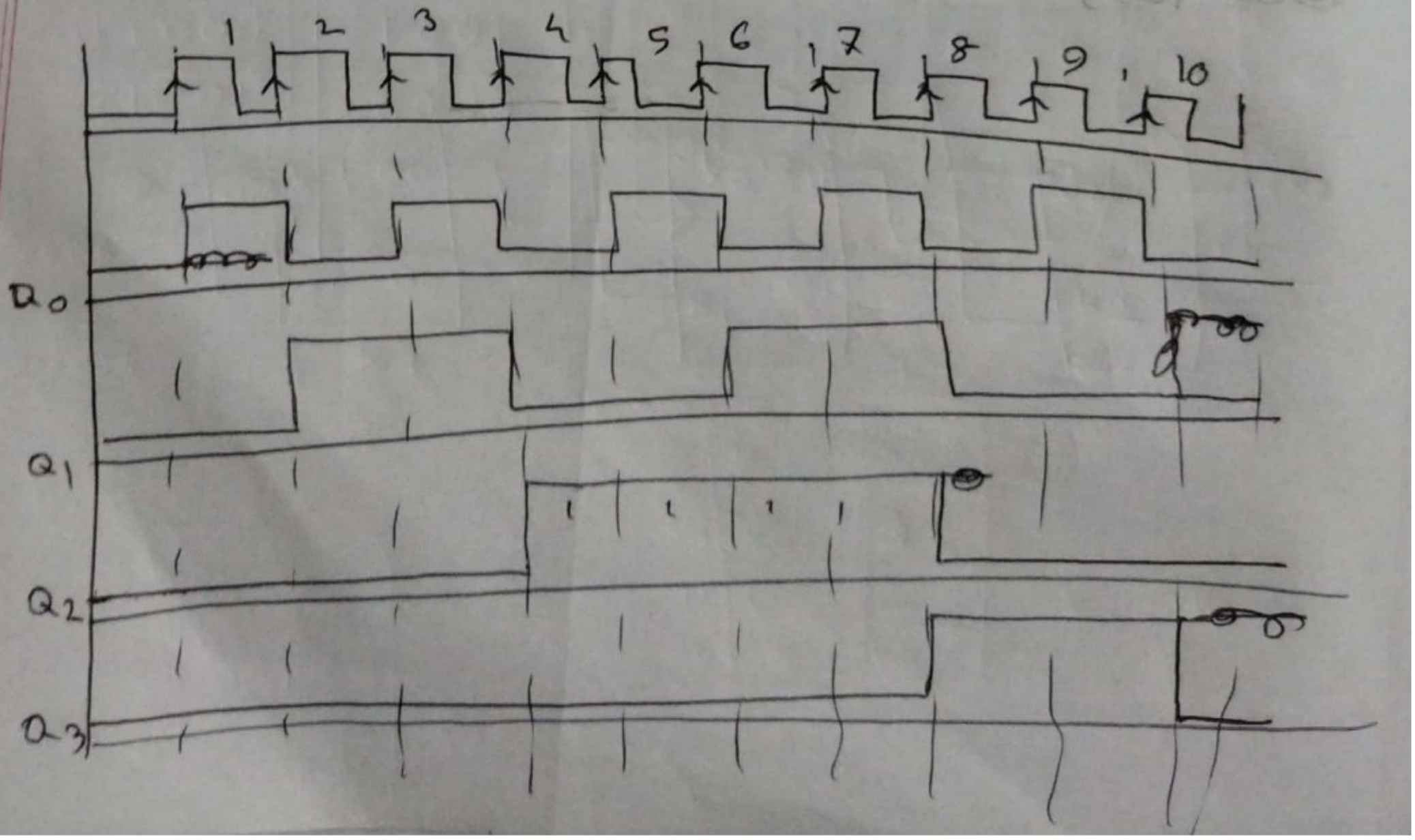
truth table 0-10

$2^3 = 8$

2^4

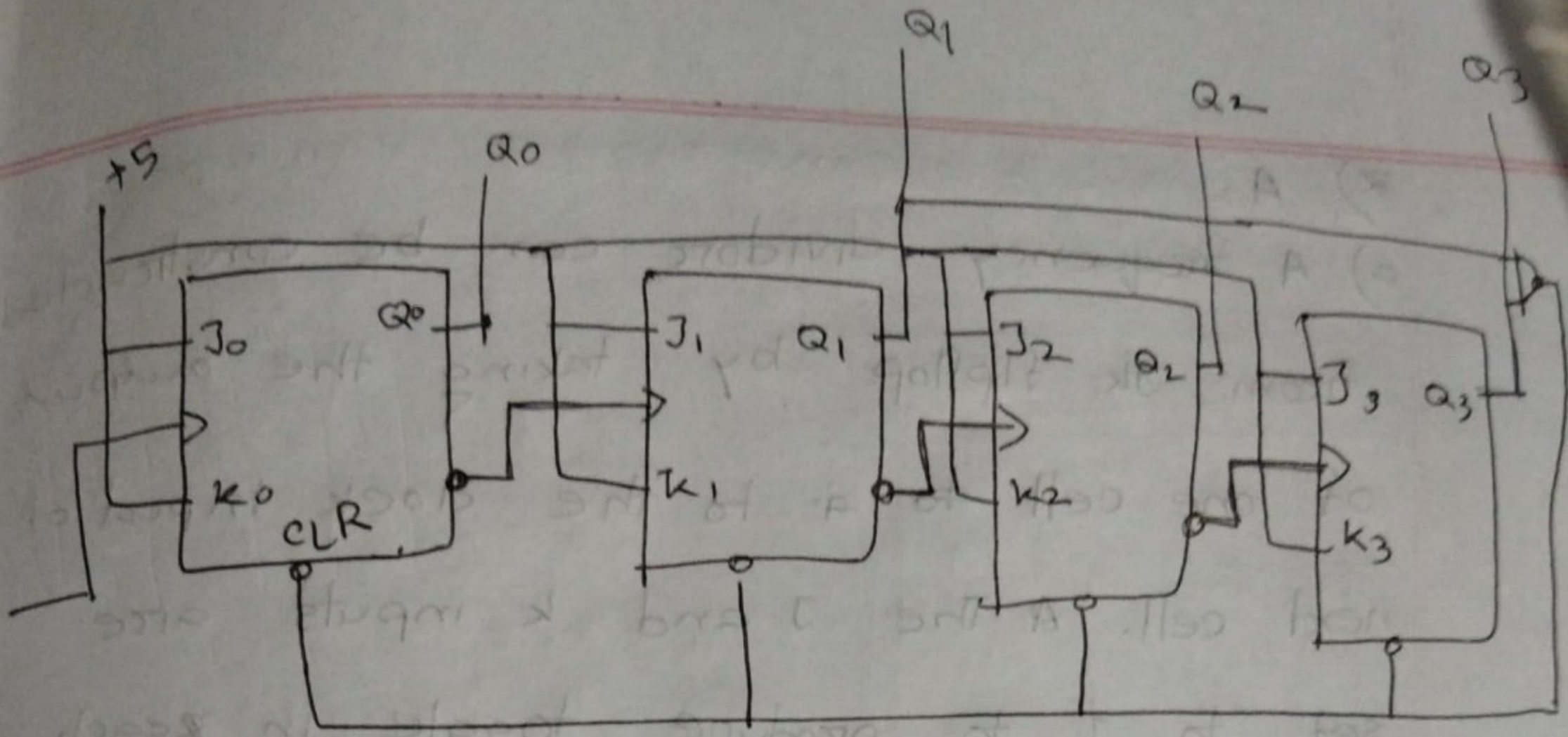
CLK	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	0	0	0	0

1001



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Asynchronous

Note → Mod 9 → $\overline{Q_0 Q_3}$ → CLR

Mod 10 → $\overline{Q_1 Q_3}$ → CLR

" 11 → $\overline{Q_1 Q_2 Q_3}$ → CLR

" 12 → $\overline{Q_2 Q_3}$ → "

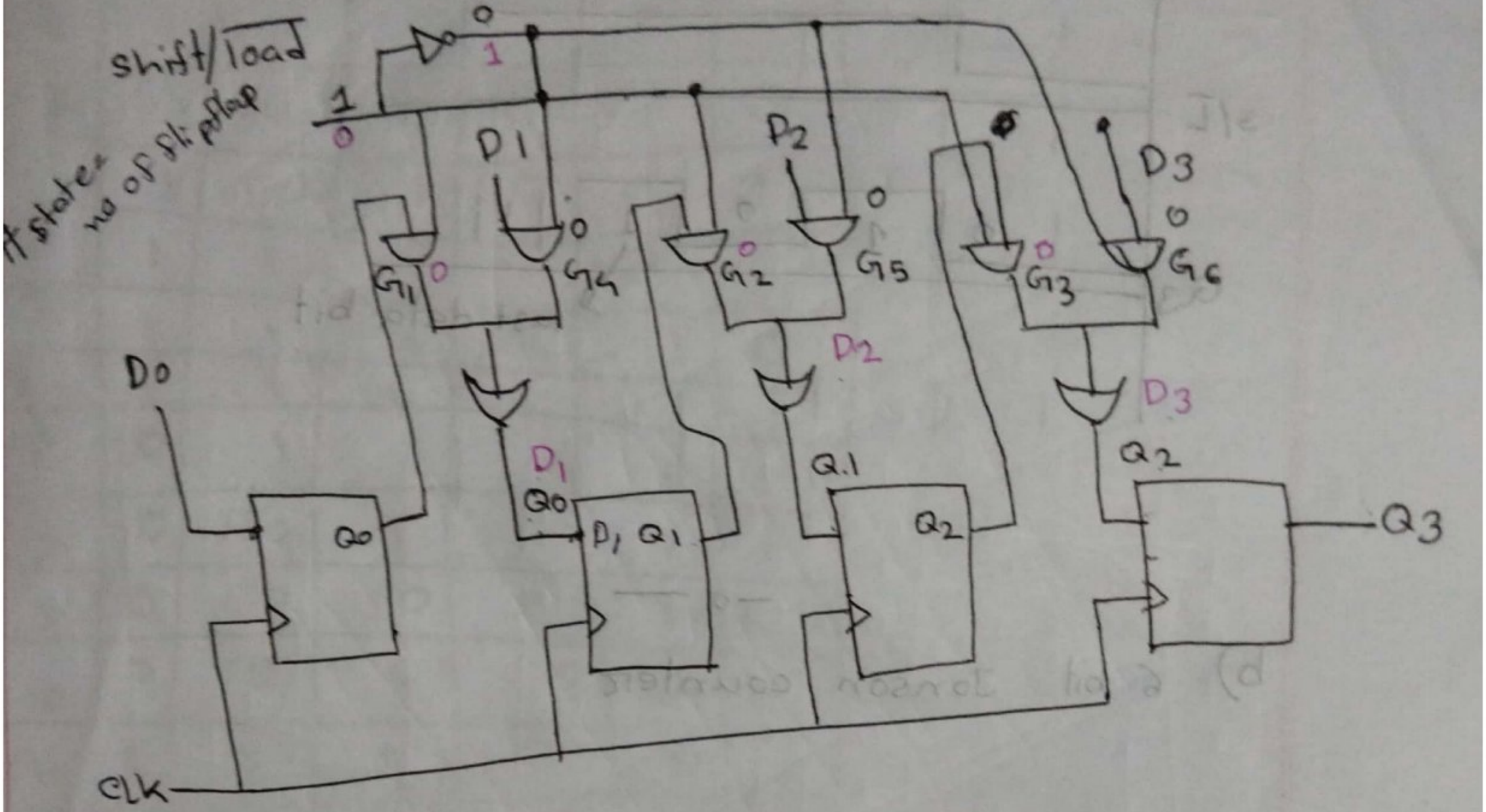
" 13 → $\overline{Q_0 Q_2 Q_3}$ → "

" 14 → $\overline{Q_1 Q_2 Q_3}$ → "

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6) a)

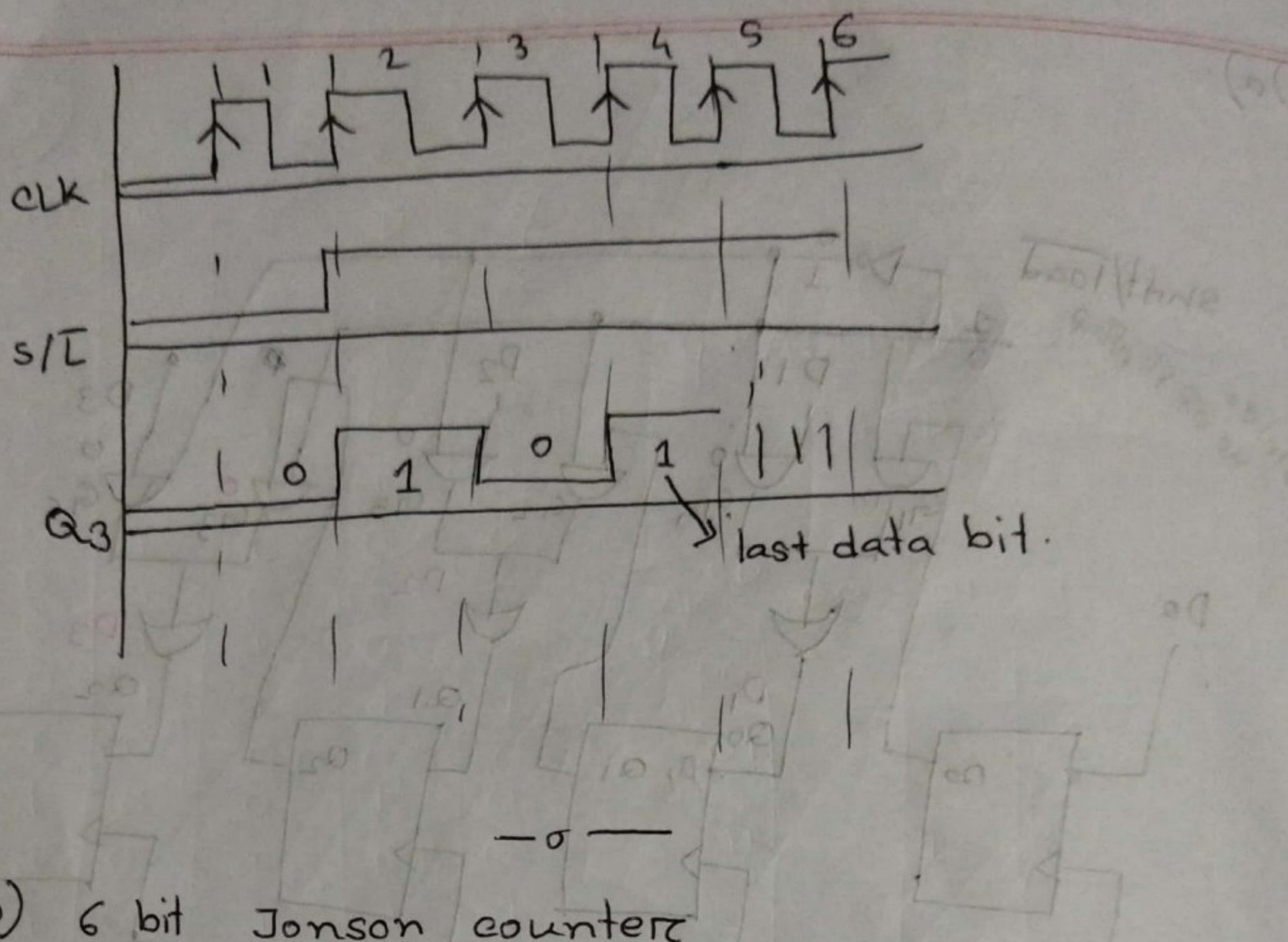


shift	CLK	D_0	D_1	D_2	D_3	Q_0	Q_1	Q_2	Q_3
0	0	1	0	1	0	0	0	0	0
0	1	1	0	1	0	1	0	1	0
1	2	1	0	1	0	1	1	0	1
1	3	1	0	1	0	1	1	1	0
1	4	1	0	1	0	1	1	1	1

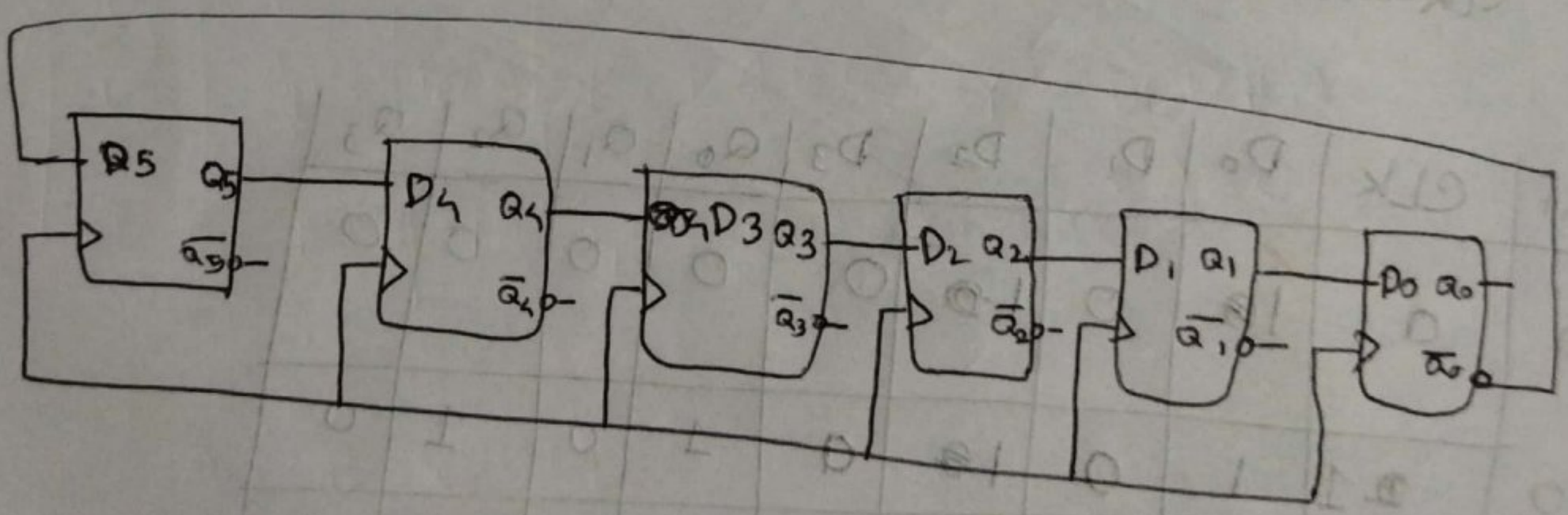
D_0 D_1 D_2 D_3
1 0 1 0

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b) 6 bit Jonson counter



Number of used states = $2n$

" " unused " = $2^n - 2n$

$n = 6$

used states = $2 \times 6 = 12$

unused " = $2^6 - 12$

$= 52$

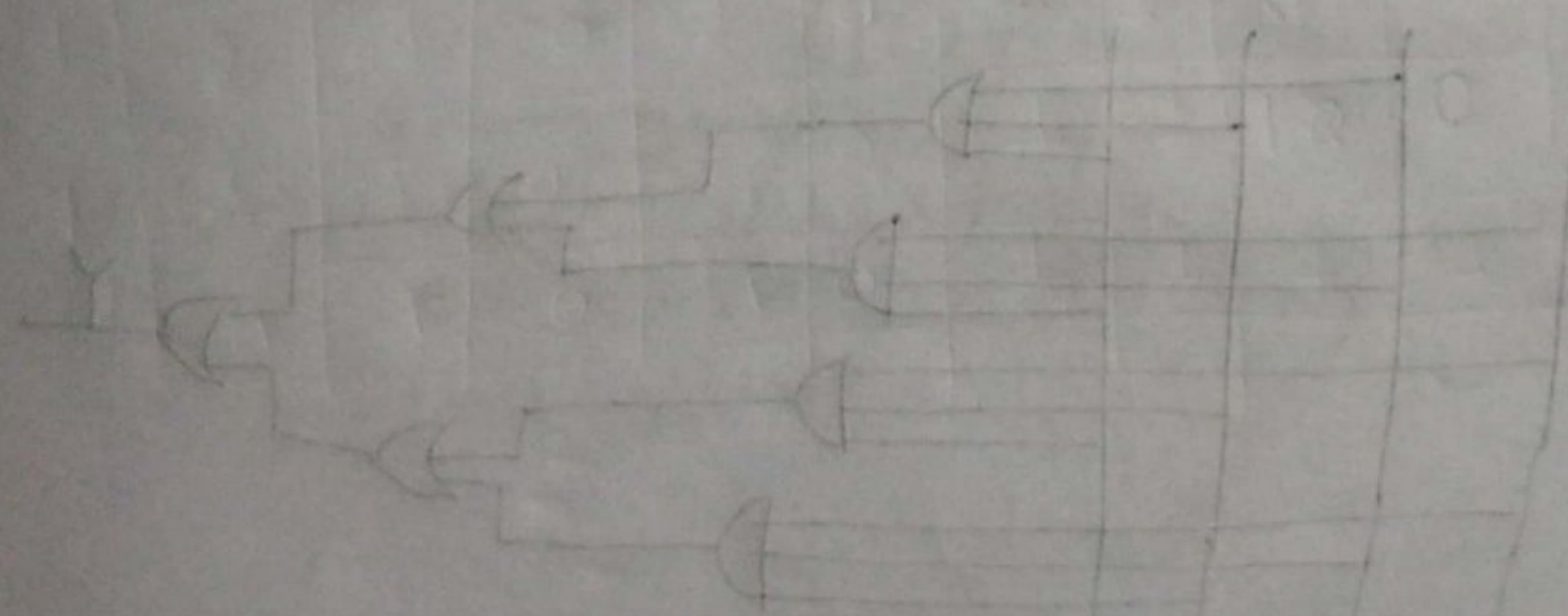
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	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0	0	0
1	1	0	0	0	0	0
2	1	1	0	0	0	0
3	1	1	1	0	0	0
4	1	1	1	1	0	0
5	1	1	1	1	1	0
6	1	1	1	1	1	1
7	0	1	1	1	1	1
8	0	0	1	1	1	1
9	0	0	0	1	1	1
10	0	0	0	0	1	1
11	0	0	0	0	0	1
12	0	0	0	0	0	0

— 6 —

Elementory comparator circuit —



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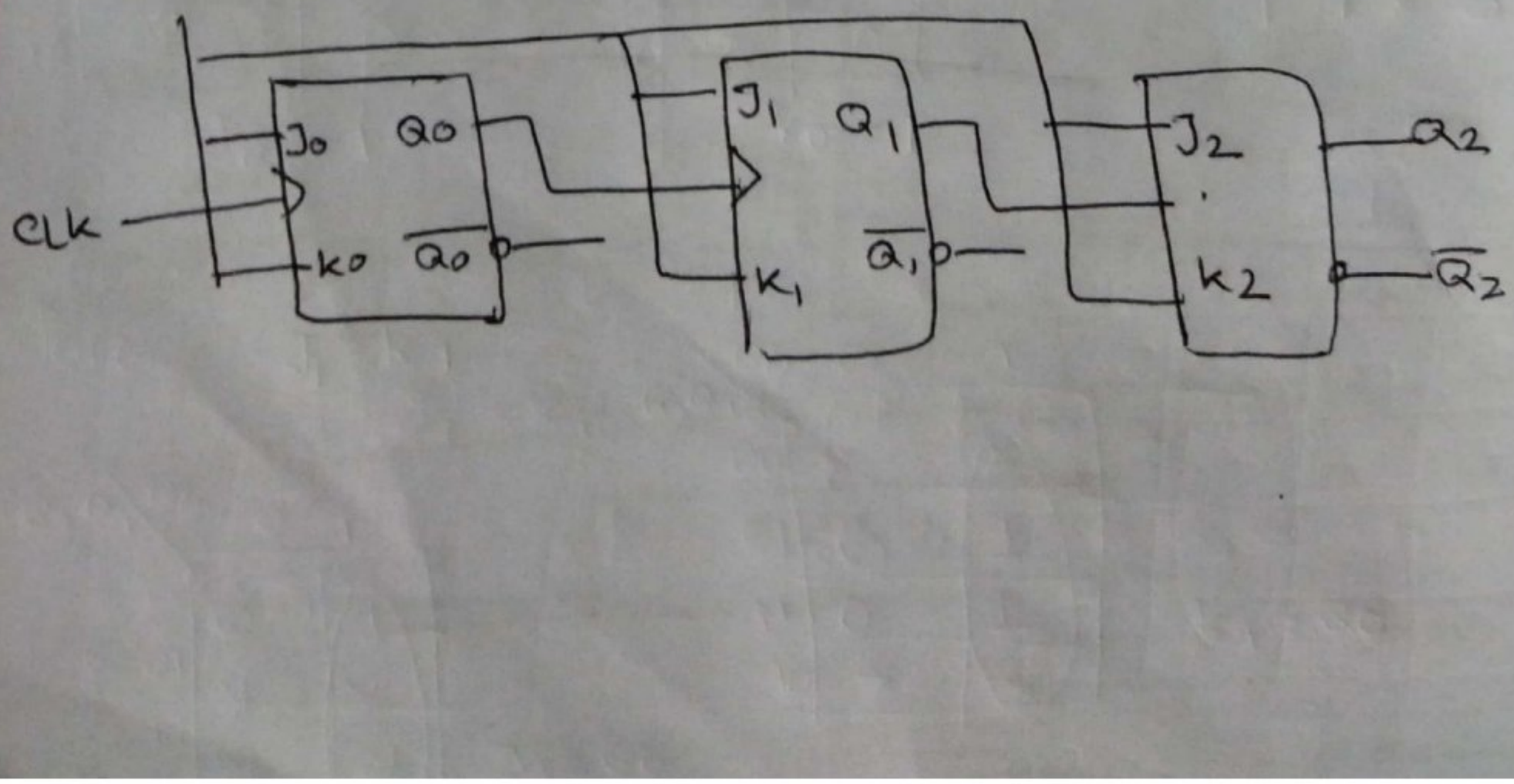
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7) A

a) A frequency dividor can be constructed from Jk flipflop by taking the output of one cell ~~to~~ ~~i~~ to the clock input of next cell. A The J and k inputs are set to 1 to produce toggle in each cycle of the clock input.

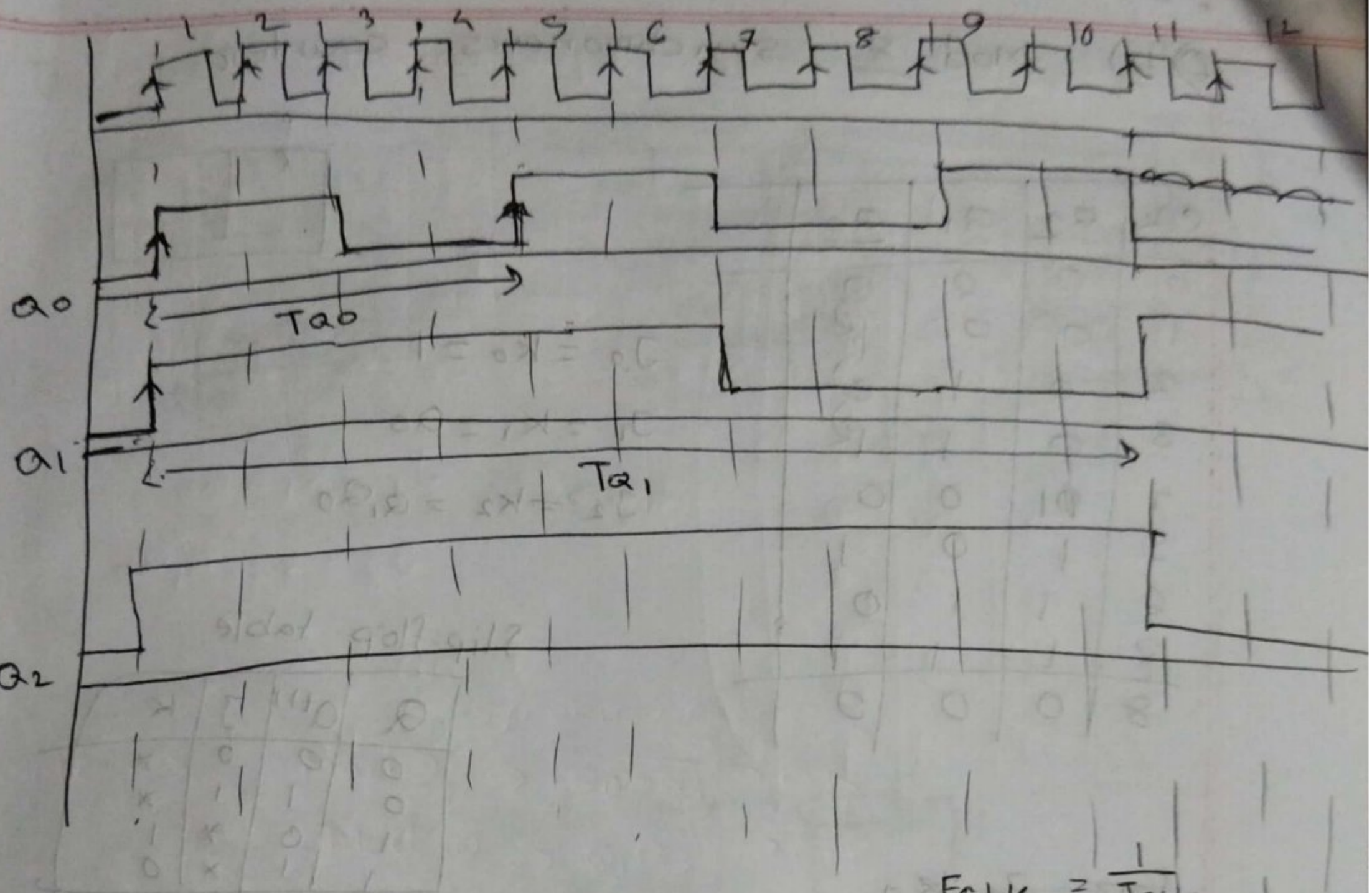
For each two toggles in first cell a toggle is produced on next cell.

So the output frequency is half of the first



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$$f_{Q_0} = \frac{1}{T_{Q_0}} = \frac{1}{2T_{CLK}} = \frac{1}{2} F_{CLK}$$

$$F_{CLK} = \frac{1}{T_{CLK}}$$

$$\frac{1}{T_{Q_0}} = \frac{1}{2F_{CLK}}$$

$$f_{Q_1} = \frac{1}{T_{Q_1}} = \frac{1}{2T_{Q_0}} = \frac{1}{2 \cdot 2T_{CLK}} = \frac{1}{4} F_{CLK}$$

$$f_{Q_2} = \frac{1}{T_{Q_2}} = \frac{1}{2T_{Q_1}} = \frac{1}{2} \cdot \frac{1}{4} F_{CLK}$$

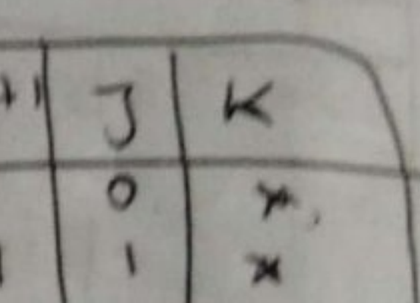
$$= \frac{1}{8} F_{CLK}.$$

J K
1 1
1 – Toggle
0

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(2) b) Mod-8 synchronous counter



clk	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

$J_0 = K_0 = 1$

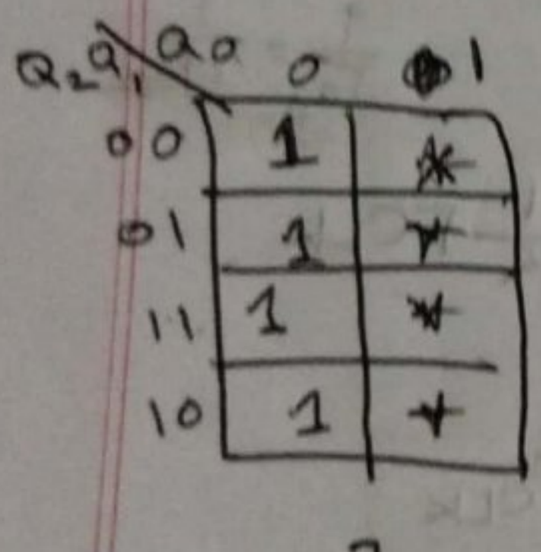
$J_1 = K_1 = Q_0$

$J_2 = K_2 = Q_1 Q_0$

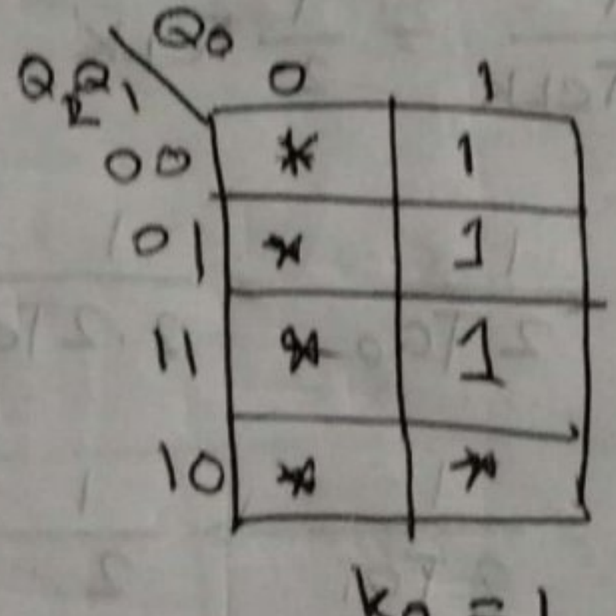
Flip-flop table

Q	Q^{t+1}	J	K
0	0	0	*
0	1	1	*
1	0	*	1
1	1	*	0

for J_0, K_0



$J_0 = 1$

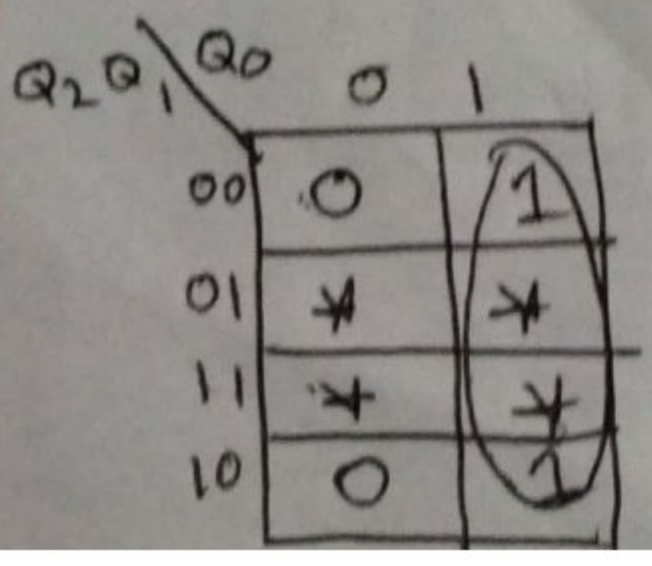


$K_0 = 1$

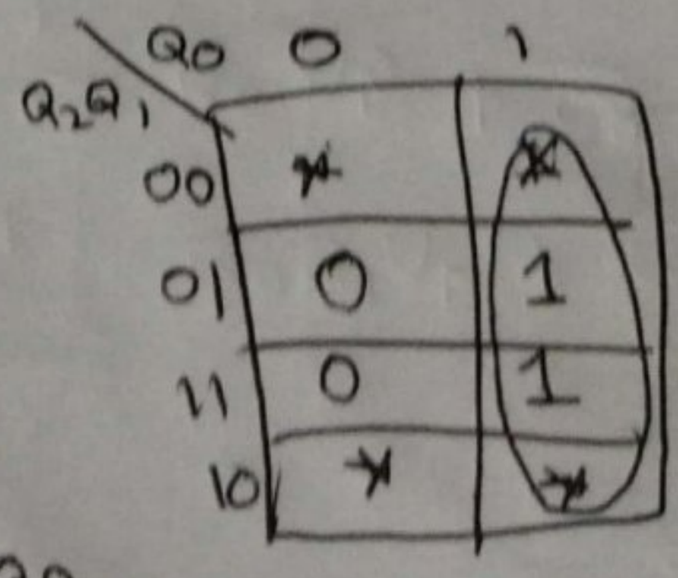
0 0 0 *
0 1 1 *
1 0 * 1
1 1 * 0

1 *
* 1

for J_1, K_1



$J_1 = Q_0$



$K_1 = Q_0$

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$8 \rightarrow 2^3$

$9 > 2^3$

J_2, K_2

Q_2Q_1 \ Q_0	0	1
00	0	0
01	0	1
11	x	x
10	x	x

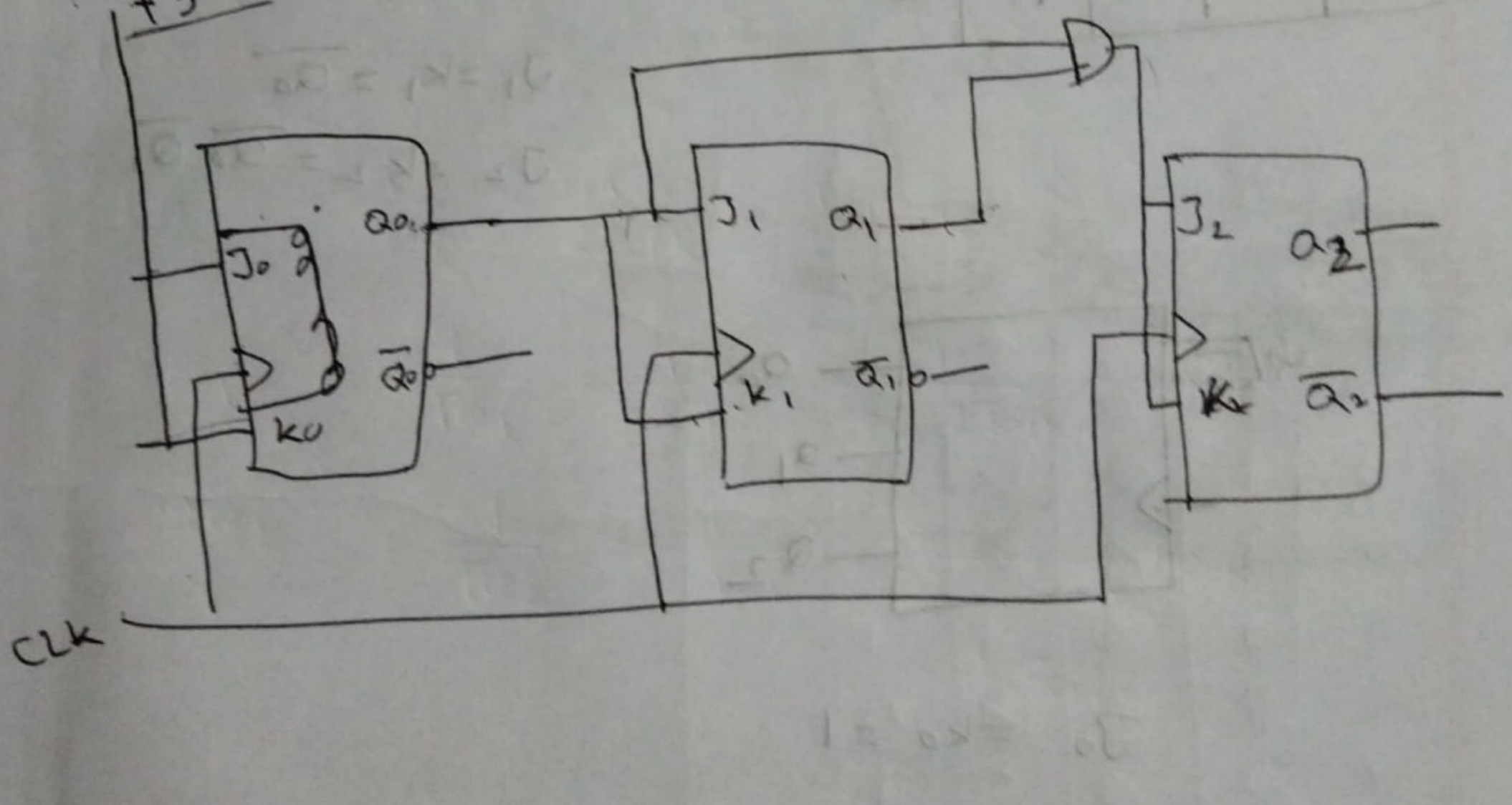
$J_2 = Q_1Q_0$

Q_2Q_1 \ Q_0	0	1
00	x	x
01	x	x
11	x	1
10	0	0

$K_1 = Q_1Q_0$

Q_2Q_1 \ Q_0	0	1
00	x	x
01	x	x
11	0	1
10	0	0

Circuit



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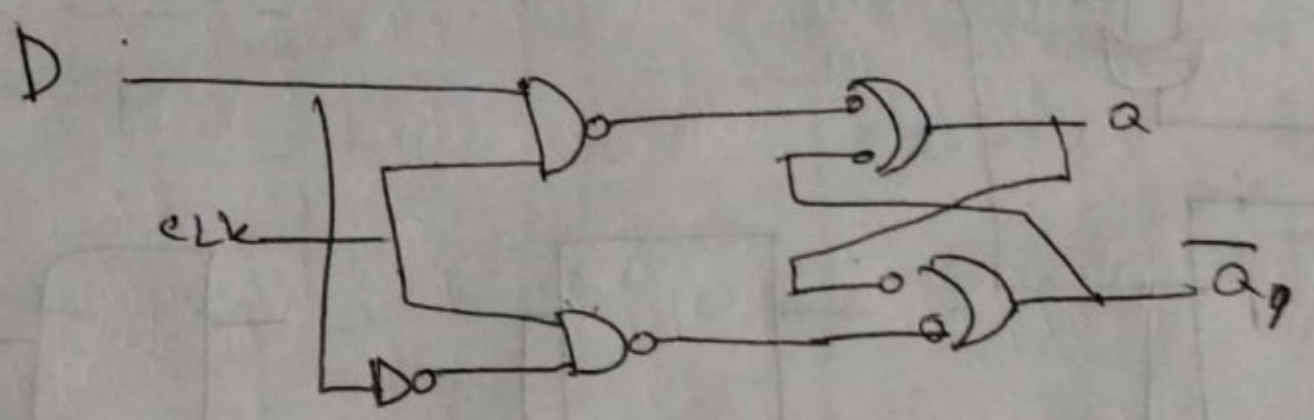
working principle

c) D flip-flop

the single input "D" is referred to as Data input.

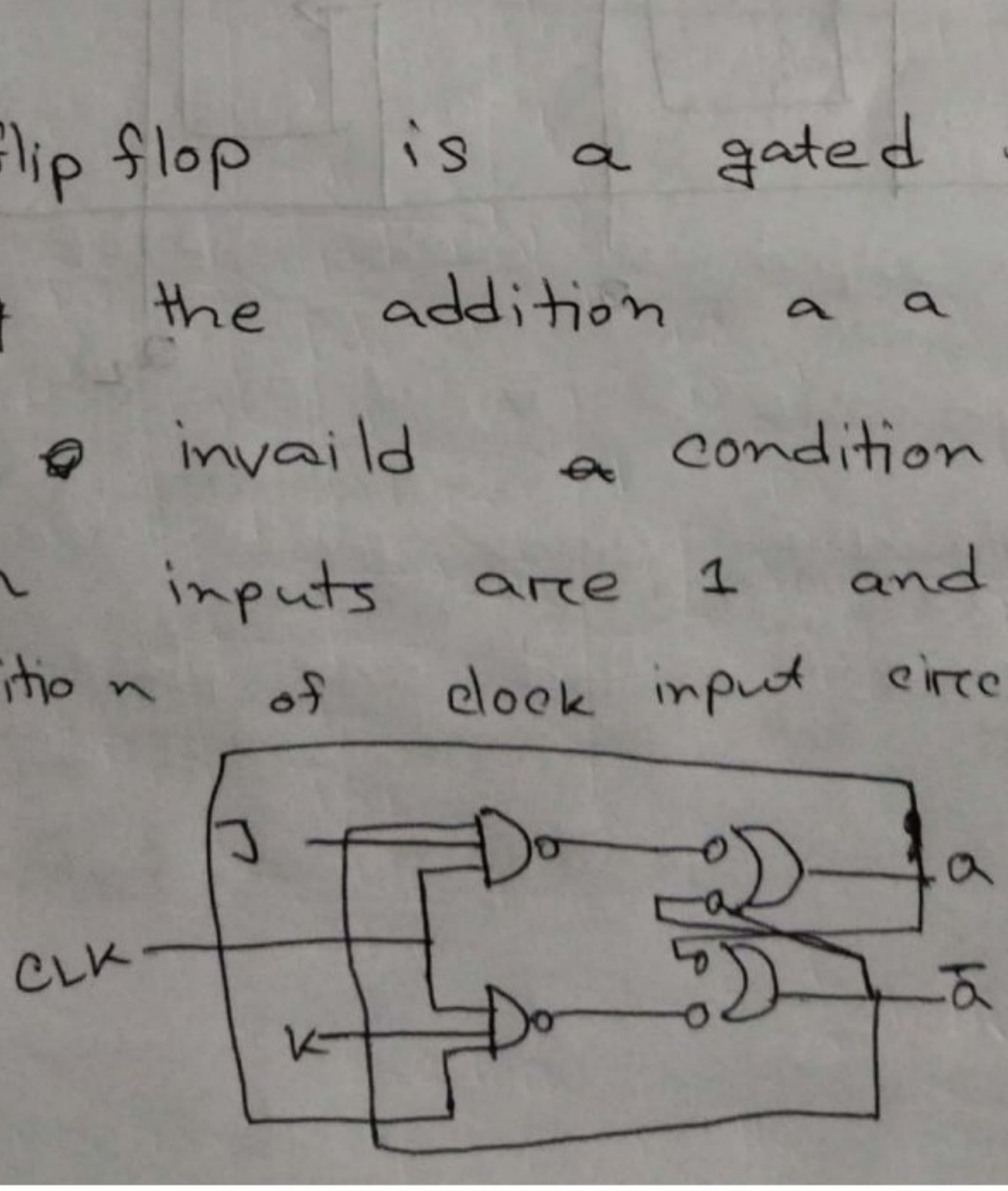
when data is set to 1 ~~flipflop~~ flop would set " " " 0 " " Reset

when clock input is set to true D input condition is copied to output Q.



Jk flip flop is a gated SR flip flop having the addition a a clock input.

The ~~o~~ invaild ~~a~~ condition occurs when both inputs are 1 and preverted by addition of clock input circuit.



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